

The OSI Model & TCP/IP Stack

The 7-Layer Model

Layer	Name	Unit	Example Devices/Protocols
7	Application	Data	HTTP, DNS, SMTP
6	Presentation	Data	TLS handshake, JPEG, SSL
5	Session	Data	NetBIOS, RPC
4	Transport	Segment	TCP, UDP
3	Network	Packet	IP, ICMP, routing
2	Data Link	Frame	Ethernet, 802.11 (Wi-Fi)
1	Physical	Bits	Cables, radio signals

You Don't Need to Memorize the Layers – You Need to THINK in Layers

Most security problems are layer confusion:

- Phishing is layer 7 (it attacks the human)
- ARP spoofing is layer 2 (forged MAC mapping)
- IP spoofing is layer 3 (forged source IP)
- SYN flood is layer 4 (TCP handshake abuse)
- Route injection is layer 3 (BGP)

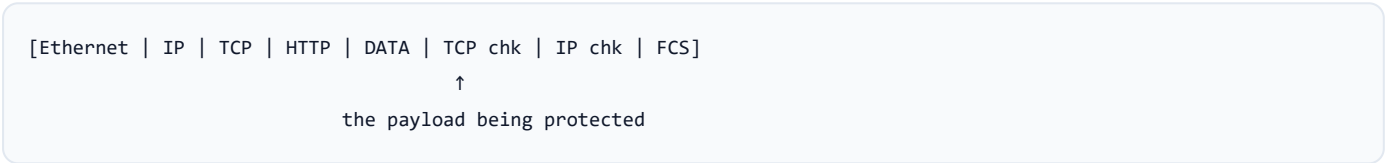
TCP/IP: The Real Stack

The OSI model is theory. The internet runs on TCP/IP – essentially 4 layers:

TCP/IP Layer	Maps To	Key Protocols
Application	L5-7	HTTP, HTTPS, DNS, SMTP, FTP, SSH
Transport	L4	TCP, UDP
Internet	L3	IP, ICMP, ARP
Link	L1-2	Ethernet, Wi-Fi

Encapsulation: The Onion

Each layer wraps the one above:



An attacker on your Wi-Fi sees the **Ethernet frame** (L2). An attacker across the internet sees the **IP packet** (L3) with source/destination IPs. An attacker on the same LAN as the target sees everything up to the application data – unless it's encrypted.

The whole reason HTTPS exists (TLS at the presentation layer) is that IP, TCP, routing, and the physical wire are all visible to middle devices. Without TLS, all of that exposed metadata includes your passwords.